Meridian Healthcare Portal | Incident Investigation Report

Training context: Completed as a guided LogNPacific cyber-range exercise, with assistance developing queries and interpreting results.

Scenario incident date: February 6, 2026 (UTC) | Investigation: 9 of 9 stages completed

Summary

I worked through the Meridian incident in Azure Log Analytics, checked the logs, and followed the attack from the first access to the final response. The attacker got into a healthcare web application, gained root-level access, and left behind a second backdoor that blended in with normal application files. The security tools blocked some activity, but they did not fully remove the threat.

What I Found

Finding	Evidence from the logs
Root-level access	A setuid-root shell, /tmp/rootbash -p, ran at 04:40:15 UTC.
Hidden backdoor	/opt/meridian/scripts/health_check ran as PID 267155 with real UID 1001 (svc_backup), effective UID 0 (root), and parent PID 1.
Command and control	Defender logged 73 blocked outbound connections to 10.1.134.57:43212, from 04:48:28 to 05:02:17 UTC.
Data at risk	The investigation identified a full export of healthcare application data, involving patient records / protected health information.
Logging damage	An early automated sweep removed sysmon.service, creating a gap in process and security event visibility.

What the Defense Tools Did

The automated sweep removed the temporary root shell, reverted altered SSH keys and crontabs, restored a tampered backup.conf, cleaned temporary directories, and removed selected services. It flagged health_check as a non-baseline file but left it running. The firewall blocked its outbound calls, which contained the network traffic but did not kill the process.

Response and What Still Needed Doing

The returning administrator captured live memory using AVML, with output saved to /tmp/evidence/memory.lime. The remaining manual response was to stop and remove the health_check backdoor and its persistence, while keeping the evidence intact. Until that happened, the incident was not fully contained.

What I Took Away

The big lesson was simple: an alert is not the same as a fix. I had to compare timestamps, check commands and permissions, separate setup-related false positives from actual attacker activity, and confirm what was blocked versus what stayed active. I also saw why automated cleanup must be checked carefully: it can miss a backdoor and take out useful logging at the same time.

Scope note: This report covers a simulated Cyber Range case and summarizes findings from the exercise logs. It does not claim a live production incident or independent confirmation that patient records left a real organization.